

# Fuel Economy Report

1974 *Motor Trend* road tests · September 2026 · prepared with tweave

**20.09 mpg**

AVERAGE FUEL ECONOMY

**-0.87**

CORRELATION, MPG VS.  
WEIGHT

**33.9 mpg**

BEST IN TEST: TOYOTA  
COROLLA

## Weight is the story

Across the 32 cars compiled for *Motor Trend* by Henderson & Velleman (1981), fuel economy falls sharply with vehicle weight. A simple linear model

$$\text{mpg}_i = \beta_0 + \beta_1 \text{weight}_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \text{Normal}(0, \sigma^2) \quad (1)$$

estimates that each additional 1,000 lbs costs about **5.34 mpg** ( $R^2 = 0.75$ ).

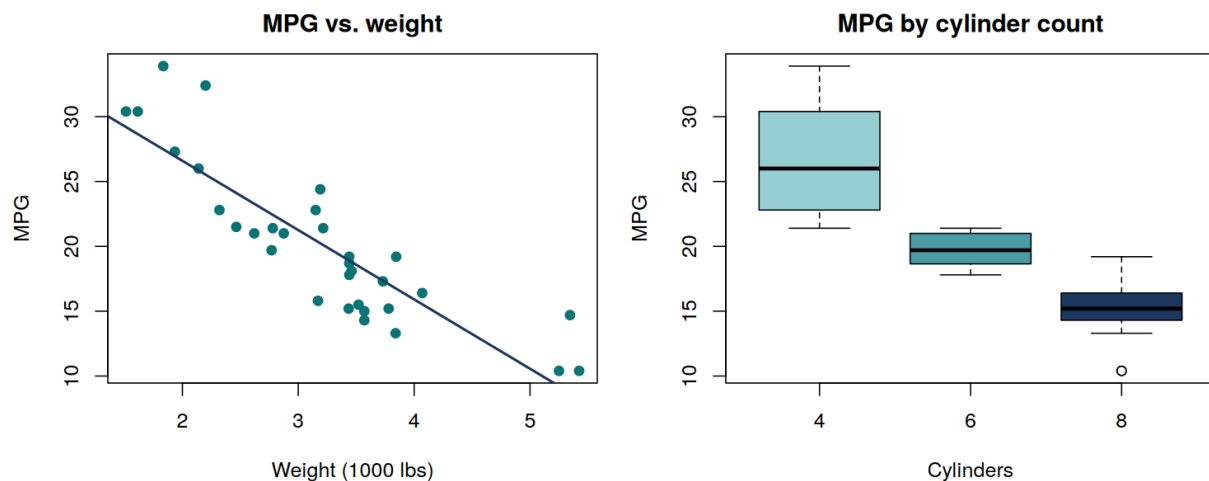


Figure 1: Fuel economy against weight (left, with fitted line) and by cylinder count (right). Heavier, higher-cylinder cars are thirstier.

As Figure 1 shows, the relationship is strong and close to linear over the observed range.

**Reading note:** the cylinder-count boxplot mostly restates the weight effect — heavier cars have more cylinders. Treat these as one finding, not two.

## Leaderboard

The efficiency podium is a clean sweep for light four-cylinder imports — every car in the top five weighs under 2,300 lbs.

| Model          | MPG  | Cylinders | Weight |
|----------------|------|-----------|--------|
| Toyota Corolla | 33.9 | 4         | 1.83   |
| Fiat 128       | 32.4 | 4         | 2.20   |
| Honda Civic    | 30.4 | 4         | 1.61   |
| Lotus Europa   | 30.4 | 4         | 1.51   |
| Fiat X1-9      | 27.3 | 4         | 1.94   |

## Does the transmission matter?

**+7.24 mpg**

Manual-transmission cars average **7.24 mpg more** than automatics in this sample (Welch  $t$ -test:  $t = 3.77$ ,  $p = 0.0014$ ).

Before crowning the stick shift, note the confound: the manual cars here are also much lighter<sup>1</sup> — transmission choice and weight travel together in 1974's lineup.

| Transmission | Mean MPG | Mean weight |
|--------------|----------|-------------|
| Automatic    | 17.1     | 3.77        |
| Manual       | 24.4     | 2.41        |

## Discussion

The efficiency leaders are all light four-cylinder imports; the sample mean of  $\bar{x} = 20.09$  mpg is dragged down by a tail of heavy V8s, several of which manage barely half that figure.

For 1974 buyers the advice writes itself: weight, not engineering exotica, predicts what you'll spend at the pump. For modern readers, the dataset remains a friendly reminder that a single well-chosen covariate often explains most of the variance. All computation was done in R (R Core Team, 2024).

## Bibliography

Henderson, H. V., & Velleman, P. F. (1981). Building multiple regression models interactively. *Biometrics*, 37(2), 391–411.

R Core Team. (2024). *R: A Language and Environment for Statistical Computing*. <https://www.r-project.org/>

<sup>1</sup>Mean weight 2.41 vs. 3.77 thousand lbs.